

7th SEMESTER – Generative AI & AI Application Development

Branch: Computer Science & Engineering

Hands-on Training Program

Duration: 60 Hours (10 Days × 6 Hours)

Training Methodology

- 20% Theory
- 80% Hands-on Practical
- Live Demonstrations
- Individual Lab Exercises
- Team-Based Activities
- Mini Project
- Daily Assignments
- Industry Best Practices

Day 1 – Introduction to Artificial Intelligence & Generative AI

Duration: 6 Hours

Topics

- Introduction to AI, Machine Learning & Deep Learning
- Evolution of Generative AI
- Large Language Models (LLMs)
- Generative AI Ecosystem
- AI Ethics & Responsible AI
- Overview of AI Development Workflow

Hands-on Lab

- Set up Python environment
- Install VS Code, Python, Jupyter Notebook
- Create your first AI-powered Python application
- Explore OpenAI, Google Gemini, and Hugging Face platforms

Assignment

Develop a simple AI-powered text generation application using a pre-trained API.

Day 2 – Prompt Engineering & AI Productivity Tools

Duration: 6 Hours

Topics

- Principles of Prompt Engineering
- Zero-shot, One-shot & Few-shot Prompting
- Chain-of-Thought Prompting
- AI Productivity Tools
- AI for Coding, Documentation & Research

Hands-on Lab

- Build effective prompts for different use cases
- Generate code, reports, presentations, and documentation
- Compare outputs from multiple AI models

Assignment

Create a prompt library for five real-world engineering scenarios.

Day 3 – Python for AI Application Development

Duration: 6 Hours

Topics

- Python Essentials for AI
- Working with APIs
- JSON Handling
- Virtual Environments
- REST API Basics

Hands-on Lab

- Connect to an AI API
- Build a Python chatbot

- Process API responses
- Handle user inputs dynamically

Assignment

Develop a console-based AI assistant.

Day 4 – Building AI-Powered Web Applications

Duration: 6 Hours

Topics

- Introduction to Flask
- Frontend Basics (HTML, CSS, JavaScript)
- Integrating AI APIs
- User Interface Design
- Secure API Key Management

Hands-on Lab

- Develop a chatbot web application
- Connect frontend with backend
- Deploy AI responses in real time

Assignment

Build an AI-powered FAQ Assistant.

Day 5 – Retrieval-Augmented Generation (RAG) & Vector Databases

Duration: 6 Hours

Topics

- Limitations of LLMs
- Introduction to RAG
- Embeddings
- Vector Databases
- Document Retrieval

Hands-on Lab

- Build a document-based chatbot
- Index PDF documents
- Query documents using natural language

Assignment

Create a college information chatbot using institutional documents.

Day 6 – AI Agents & Workflow Automation

Duration: 6 Hours

Topics

- AI Agents
- Tool Calling
- Function Calling
- Multi-Agent Systems
- AI Workflow Automation

Hands-on Lab

- Build an AI scheduling assistant
- Integrate external APIs
- Automate repetitive workflows

Assignment

Develop an AI agent for task management.

Day 7 – Multimodal AI Applications

Duration: 6 Hours

Topics

- Image Generation
- Speech-to-Text
- Text-to-Speech
- OCR

- Vision Models

Hands-on Lab

- Build an image captioning application
- Generate images from text prompts
- Convert speech into text
- Create a voice-enabled chatbot

Assignment

Develop a multimodal AI application integrating text, image, or voice.

Day 8 – AI Deployment & Cloud Integration

Duration: 6 Hours

Topics

- Model Deployment
- API Deployment
- Docker Basics
- Cloud AI Services
- Performance Monitoring

Hands-on Lab

- Deploy an AI web application
- Containerize using Docker
- Host the application on a cloud platform

Assignment

Publish an AI application with public access.

Day 9 – Team-Based AI Project Development

Duration: 6 Hours

Activity

Students work in teams to build an AI-powered application.

Suggested Projects

- AI Resume Analyzer
- AI Interview Preparation Assistant
- AI Study Companion
- AI Code Reviewer
- AI Content Generator
- AI Healthcare Assistant
- AI Attendance Assistant
- AI Customer Support Bot

Trainer Evaluation

- Design
- AI Integration
- Functionality
- Innovation
- Team Collaboration

Day 10 – Capstone Project, Presentation & Assessment

Duration: 6 Hours

Capstone Project

Each team presents a complete AI-powered application addressing a real-world problem.

Practical Assessment

Each student must:

- Design prompts for an AI use case
- Integrate an LLM API into an application
- Build a web-based AI application
- Implement a RAG workflow with a document knowledge base
- Demonstrate an AI agent or automation feature
- Deploy the application and present its functionality
- Document the project with installation, usage, and architecture details

Final Evaluation

- Prompt Engineering Skills
- Python Programming
- AI API Integration
- Web Application Development
- RAG Implementation
- Deployment & Documentation
- Team Collaboration
- Innovation & Problem Solving

Recommended Tools & Technologies

- **Programming:** Python 3.x
- **IDE:** VS Code, Jupyter Notebook
- **Frameworks:** Flask, FastAPI (Introduction)
- **AI Platforms:** OpenAI API, Google Gemini API, Hugging Face
- **Libraries:** LangChain, LlamaIndex, Transformers
- **Vector Databases:** ChromaDB, FAISS
- **Databases:** SQLite (Basics)
- **Deployment:** Docker, Render, Railway, Hugging Face Spaces
- **Version Control:** Git & GitHub

Learning Outcomes

By the end of this 60-hour program, students will be able to:

- Understand the fundamentals of Artificial Intelligence, Generative AI, and Large Language Models (LLMs).
- Apply prompt engineering techniques to solve real-world engineering and business problems.
- Develop AI-powered applications using Python and REST APIs.
- Build interactive web applications integrated with Generative AI services.

- Implement Retrieval-Augmented Generation (RAG) systems using vector databases and document retrieval.
- Create AI agents capable of automating tasks and interacting with external tools.
- Develop multimodal AI applications that process text, images, and speech.
- Deploy AI applications using modern cloud platforms and containerization tools.
- Collaborate in teams to design, develop, test, and present end-to-end AI solutions.
- Demonstrate industry-ready skills in AI application development, deployment, and responsible AI practices.

